

Project Title

Air Quality Pattern Discovery and Pollution Anomaly Detection Using Unsupervised Machine Learning

Real-World Problem

Air pollution is a major environmental and public-health concern in the Ghaziabad–NCR region. Air-quality levels can change significantly due to traffic, industrial emissions, construction dust, weather conditions, seasonal changes, and events such as stubble burning or festivals.

Although air-quality data is available through monitoring stations, it can be difficult to manually identify recurring pollution patterns, classify days with similar air quality, and detect unusually severe pollution events. Traditional analysis often depends on predefined labels such as “good,” “moderate,” or “severe,” but many hidden patterns may remain unnoticed.

This project uses unsupervised machine learning to automatically group similar air-quality observations and identify abnormal pollution events without requiring pre-labeled categories.

Problem Statement

Develop an unsupervised machine learning system that analyzes air-quality and weather data to discover hidden pollution patterns and detect unusual high-pollution days or locations.

The system will use pollutant variables such as **PM2.5, PM10, NO₂, CO**, along with weather-related variables such as **temperature, humidity, and wind speed**. It will cluster similar observations using K-Means and identify pollution anomalies using Isolation Forest.

Project Objective

The main objective of this project is to analyze air-quality data and identify meaningful pollution patterns using unsupervised learning techniques.

The specific objectives are:

- To collect and preprocess air-quality and weather data for Ghaziabad/NCR.
- To explore relationships between pollutants and weather conditions.
- To use **K-Means Clustering** to group days or locations with similar air-quality characteristics.
- To use **PCA (Principal Component Analysis)** to reduce dimensions and visualize clusters.
- To use **Isolation Forest** to detect unusual or highly polluted observations.
- To evaluate clustering quality using metrics such as **Silhouette Score** and **Davies–Bouldin Score**.

- To generate insights that can help identify high-risk pollution periods and support awareness or planning efforts.

Expected Outcome

The project will classify air-quality observations into clusters such as relatively clean days, moderate-pollution days, high-pollution days, and weather-driven pollution patterns. It will also flag unusual observations, such as sudden spikes in PM2.5 or PM10, as potential pollution anomalies.

The final output can include graphs, PCA cluster plots, pollutant trends, anomaly charts, and a simple dashboard showing pollution categories and detected abnormal days.

DATASET SELECTION:

Air Quality Data in India (2015 - 2020)

The dataset contains air quality data and AQI (Air Quality Index) at hourly and daily level of various stations across multiple cities in India.

Cities

Ahmedabad, Aizawl, Amaravati, Amritsar, Bengaluru, Bhopal, Brajrajnagar, Chandigarh, Chennai, Coimbatore, Delhi, Ernakulam, Gurugram, Guwahati, Hyderabad, Jaipur, Jorapokhar, Kochi, Kolkata, Lucknow, Mumbai, Patna, Shillong, Talcher, Thiruvananthapuram, Visakhapatnam

Dataset link:

<http://kaggle.com/datasets/rohanrao/air-quality-data-in-india>

3. Data Preprocessing

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Data preprocessing is required to convert the raw air-quality dataset into a clean and suitable form for unsupervised machine learning. Since clustering algorithms such as K-Means and anomaly-detection models such as Isolation Forest are sensitive to missing values, duplicate records, and different feature scales, the following preprocessing steps will be performed.

Step 1: Load and Understand the Dataset

The `city_day.csv` dataset will be loaded using Python libraries such as Pandas. The dataset will be inspected using functions such as `head()`, `info()`, `describe()`, and `isnull().sum()`.

The structure, data types, number of records, numerical columns, categorical columns, and missing-value counts will be identified.

Expected Outcome:

A clear understanding of the dataset structure and identification of relevant columns for the project.

Step 2: Select Relevant Records

The dataset contains air-quality data for multiple Indian cities. For this project, records related to Delhi/NCR will be selected where available. If sufficient Ghaziabad records are not present, Delhi data or multiple NCR cities will be used for analysis.

The selected data will include the `City` and `Date` columns along with pollutant measurements such as `PM2.5`, `PM10`, `NO2`, and `CO`. Optional pollutant features such as `SO2`, `O3`, and `NH3` may also be included if they have sufficient valid values.

Expected Outcome:

A focused dataset containing air-quality observations relevant to the Delhi/NCR pollution context.

Step 3: Convert Date Column and Create Time-Based Features

The `Date` column will be converted into datetime format. New features such as month, year, season, day of week, and winter/summer indicator may be extracted.

These features can help identify seasonal pollution patterns. For example, winter months may show higher particulate pollution due to lower wind speed, temperature inversion, and other environmental factors.

Expected Outcome:

A dataset enriched with time-based information for trend analysis and seasonal interpretation.

Step 4: Handle Missing Values

The dataset may contain missing values in pollutant columns because monitoring stations may not record every pollutant every day.

Columns with a very high percentage of missing values will be removed if they are not essential. For important pollutant columns such as `PM2.5`, `PM10`, `N02`, and `CO`, missing values will be handled using median imputation. Median imputation is suitable because pollution values can contain extreme spikes, and the median is less affected by outliers than the mean.

If a record has missing values in too many important columns, that record will be removed.

Expected Outcome:

A complete dataset without null values in the selected model features.

Step 5: Remove Duplicate Records

Duplicate rows will be checked using the `duplicated()` function. Exact duplicate records will be removed.

Duplicates may occur due to repeated data collection, merging of files, or repeated uploads. Keeping duplicate observations can distort cluster sizes and affect anomaly detection.

Expected Outcome:

A dataset in which every row represents a unique daily air-quality observation.

Step 6: Handle Invalid and Extreme Values

Pollutant values will be checked for invalid values such as negative concentrations, zero values where scientifically unrealistic, or extremely large values caused by data-entry errors.

Invalid records will either be removed or replaced with missing values and then imputed. Extreme but valid pollution spikes will not automatically be removed because they may represent real high-pollution events and are important for anomaly detection.

Expected Outcome:

Data-entry errors are removed while meaningful pollution anomalies are preserved.

Step 7: Feature Encoding

The `City` column is categorical. If multiple cities are included in the analysis, it will be converted into numerical form using One-Hot Encoding.

The `Date` column will not be used directly in clustering because it is not a numerical pollution measurement. Instead, extracted features such as month or season may be used if required.

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The **AQI_Bucket** column will not be used as a model input because it is already a predefined pollution category. It may be used later only to interpret whether the discovered clusters are meaningful.

Expected Outcome:

All model-input features are numerical and suitable for machine learning algorithms.

Step 8: Select Final Features

The main features selected for clustering and anomaly detection will be:

- PM2.5
- PM10
- NO2
- CO
- SO2, O3, and NH3, if sufficient data is available
- Month or season, if seasonal effects are included
- City-encoded columns, if multiple cities are included

The **AQI** and **AQI_Bucket** columns will not be used for training because the project aims to discover pollution patterns without relying on predefined labels.

Expected Outcome:

A final feature matrix containing only meaningful numerical variables for unsupervised learning.

Step 9: Feature Scaling and Normalization

Pollutant features have different units and value ranges. For example, PM2.5 may have much larger values than CO. Therefore, StandardScaler will be used to standardize all selected numerical features.

Standardization transforms features so that they have a mean close to zero and a standard deviation close to one. This is important because K-Means uses distance calculations and may otherwise give more importance to features with larger values.

Expected Outcome:

All features contribute fairly to clustering and anomaly detection.

Step 10: Exploratory Data Analysis

Exploratory Data Analysis will be performed before applying machine learning models. The following visualizations will be created:

- Missing-value heatmap or bar chart
- Histograms of PM2.5, PM10, NO2, and CO

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- Boxplots to identify pollutant distributions and extreme values
- Line charts showing pollution trends over time
- Monthly or seasonal average pollution charts
- Correlation heatmap between pollutants and weather-related variables, if available
- City-wise pollution comparison charts, if multiple cities are used

EDA will help understand the distribution of pollutants, identify seasonal trends, observe relationships between variables, and detect possible data-quality issues.

Expected Outcome:

Clear insights into pollution behavior, pollutant relationships, seasonal variation, and potential outliers before model development.

Final Preprocessing Output

After preprocessing, the final dataset will be clean, free from duplicate records, complete for selected features, numerically encoded, and standardized. This prepared dataset will be used as input for PCA visualization, K-Means clustering, and Isolation Forest anomaly detection.

MODEL DEVELOPMENT

For this project, K-Means Clustering will be used as the main and only unsupervised machine learning model.

K-Means is suitable because the objective of the project is to identify hidden air-quality patterns by grouping similar observations based on pollutant concentration levels. The dataset does not contain predefined labels for pollution patterns, so K-Means will automatically divide the observations into meaningful clusters.

The model will use standardized numerical features such as PM2.5, PM10, NO2, CO, SO2, O3, and NH3, depending on the availability of valid values in the dataset. Each row in the dataset will represent one daily air-quality observation for a selected city or location.

Working of K-Means Clustering

K-Means divides the dataset into a predefined number of clusters, represented by K. The model calculates the similarity between observations using distance. Observations with similar pollutant values are placed in the same cluster.

For example, the model may identify groups such as:

- Low-pollution days with low PM2.5 and PM10 values
- Moderate-pollution days with average pollutant levels
- High-pollution days with high particulate matter values
- Severe-pollution days with high levels of multiple pollutants

The exact number and meaning of clusters will be decided after training and analyzing the results.

Selecting the Optimal Number of Clusters

Before training the final K-Means model, different values of K, such as 2 to 10, will be tested.

The optimal number of clusters will be selected using the following methods:

- **Elbow Method:** The Within-Cluster Sum of Squares

(WCSS) will be calculated for different K values. The

point where the decrease in WCSS

becomes gradual will indicate the possible optimal K value.

- **Silhouette Score:** The Silhouette Score will measure how well observations belong to their assigned clusters. A higher Silhouette Score indicates that the clusters are more distinct and well separated.

The final value of K will be selected by considering the Elbow Method, Silhouette Score, and the interpretability of the resulting pollution groups.

Training the Model

After selecting the optimal K value, the K-Means model will be trained using the scaled air-quality dataset.

The model will assign a cluster label to every observation. These labels will be stored in a new column named **Cluster**. The average pollutant values of each cluster will be calculated to understand the pollution characteristics represented by that cluster.

PCA for Visualization

Principal Component Analysis (PCA) will be used only to visualize the K-Means clusters in two dimensions.

Since the dataset contains multiple pollutant features, PCA will reduce them into two principal components. A scatter plot will then be created using these two components, where each point represents an air-quality observation and the color of the point represents its assigned K-Means cluster.

PCA will not be used as a separate machine learning model. It will only help in visually understanding the separation between clusters.

Expected Output

The K-Means model will generate:

- A cluster label for each air-quality observation

- The optimal number of clusters
- Cluster centroids representing average pollutant levels
- Pollution-pattern groups such as low, moderate, high, and severe pollution
- Elbow Method graph
- Silhouette Score results
- PCA-based cluster visualization

The final model will help identify hidden pollution patterns in the selected air-quality dataset without using predefined labels.

Conclusion

This project aims to analyze air-quality data and discover hidden pollution patterns using K-Means Clustering, an unsupervised machine learning algorithm. The dataset will include important pollutant measurements such as PM2.5, PM10, NO2, CO, SO2, O3, and NH3 collected from Delhi/NCR or other relevant Indian cities.

The data will first be cleaned by handling missing values, removing duplicate records, selecting relevant features, and scaling numerical values. After preprocessing, K-Means Clustering will group similar air-quality observations without using predefined pollution labels.

The generated clusters are expected to represent different pollution conditions, such as low-pollution days, moderate-pollution days, high-particulate pollution days, and

severe-pollution days. The optimal number of clusters will be selected using the Elbow Method and Silhouette Score. PCA will be used to visualize the clusters in two dimensions and understand how well different pollution patterns are separated.

The project will help identify meaningful air-quality trends and pollution groups from raw environmental data. These insights can support public awareness, environmental planning, and better understanding of pollution behavior in the Delhi/NCR region. The project also demonstrates the complete unsupervised machine learning workflow, including dataset selection, preprocessing, model development, evaluation, visualization, and interpretation of results.

1. Semantic Segmentation vs Instance Segmentation vs Panoptic Segmentation

Semantic Segmentation classifies every pixel in an image into a category. All objects belonging to the same category are given the same label. For example, if an image contains three cars, all car pixels will be labelled as “car,” but the model will not distinguish between Car 1, Car 2, and Car 3.

Instance Segmentation also classifies every pixel, but it separates different objects of the same category. For example, if an image contains three cars, each car receives a separate identity, such as Car 1, Car 2, and Car 3. It is useful when identifying individual objects is important.

Panoptic Segmentation combines semantic and instance segmentation. It assigns a class label to every pixel and also separates individual objects where needed. For example, road and sky are labelled as background regions, while each person and car is identified as a separate object instance.

Type	Pixel Classification	Separates Same-Type Objects?	Example
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Semantic Segmentation	Yes	No	All cars are labelled “car”
Instance Segmentation	Yes	Yes	Each car gets a different label
Panoptic Segmentation	Yes	Yes, for object instances	Road is one region; each car and person is separate

2. What is Fuzzy Logic? How Is It Different from Boolean Logic?

Fuzzy Logic is a method of reasoning in which a value can have a degree of truth between 0 and 1. Instead of saying something is completely true or completely false, fuzzy logic allows partial truth.

For example, in a temperature-control system, a temperature of 30°C may be considered “warm” with a membership value of 0.7 and “hot” with a membership value of 0.3. This makes fuzzy logic useful for real-world situations where categories are not always exact.

Boolean Logic works only with two fixed values:

- 0 or False
- 1 or True

For example, in Boolean Logic, a room is either “hot” or “not hot.” There is no middle value.

Feature	Boolean Logic	Fuzzy Logic
Truth values	Only 0 or 1	Any value between 0 and 1
Decision type	Exact and fixed	Gradual and flexible

Example	Temperature is hot / not hot	Temperature is slightly hot, warm, or very hot
Best use	Digital circuits, exact conditions	Washing machines, AC systems, traffic control, decision systems

3. Customer Dataset Without Labels

Suppose a dataset contains customer **Age**, **Annual Income**, and **Spending Score**, but no target labels.

1. Is this supervised or unsupervised learning?

This is an **unsupervised learning** problem because no output label or target variable is provided. The aim is to discover hidden patterns or groups of similar customers based on their age, income, and spending behavior.

2. Which clustering algorithm would you choose first and why?

The first algorithm to use would be **K-Means Clustering**.

K-Means is suitable because the dataset contains numerical features and the objective is to group similar customers into segments. It is simple, fast, easy to implement, and easy to explain in a report. It can create useful customer groups such as:

- Young customers with high spending scores
- High-income customers with low spending scores
- Low-income customers with moderate spending scores
- Older customers with stable spending patterns

Before applying K-Means, the features should be scaled using StandardScaler because Annual Income, Age, and Spending Score may have different ranges.

The Elbow Method and Silhouette Score can be used to choose the most suitable number of clusters, represented by **K**.

3. What assumptions does K-Means make about the data?

K-Means makes the following assumptions:

- The data can be divided into a predefined number of clusters, **K**.
- Each cluster is roughly spherical or circular in shape.

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- Customers within the same cluster are closer to each other than to customers in other clusters.
- Clusters are separated mainly based on distance, usually Euclidean distance.
- Features should be on a similar scale; otherwise, a large-range feature such as Annual Income may dominate the clustering result.
- Clusters are expected to have somewhat similar size and density.
- Extreme outliers can affect cluster centroids and reduce clustering quality.

Therefore, K-Means is a strong first choice for this customer segmentation dataset, but the data should be scaled and checked for outliers before training the model.